Assignment 8:-

1.Write a query that uses a subquery to obtain all orders for the customer named Cisneros. Assume you do not know his customer number (cnum).

```
d2_96712_Devanand> SELECT * FROM orders WHERE cnum = (SELECT cnum
 FROM customers WHERE cname = 'Cisneros');
+------+---------+------------+------+------+
| onum | amt     | odate      | cnum | snum |
+------+---------+------------+------+------+
| 3001 |   18.69 | 1990-10-03 | 2008 | 1007 |
| 3006 | 1098.16 | 1990-10-03 | 2008 | 1007 |
+------+---------+------------+------+------+
2 rows in set (0.178 sec)
```

2.Write a query to display the order amount, customer names and ratings of all customers where the amount is greater than the average amount from the orders table.

```
d2_96712_Devanand> SELECT o.amt, c.cname, c.rating
    -> FROM Orders o
    -> JOIN Customers c ON o.cnum = c.cnum
    -> WHERE o.amt > (
    ->     SELECT AVG(amt)
    ->     FROM Orders
    -> );
+---------+---------+--------+
| amt     | cname   | rating |
+---------+---------+--------+
| 5160.45 | Liu     |    200 |
| 9891.88 | Clemens |    100 |
| 4723.00 | Clemens |    100 |
+---------+---------+--------+
3 rows in set (0.112 sec)
```

3. Write a query that selects the total amount in orders for each salesperson for whom this total is greater than the amount of the largest order in the table.

```
d2_96712_Devanand> SELECT snum, SUM(amt) AS total_sales
    -> FROM Orders
    -> GROUP BY snum
    -> HAVING SUM(amt) > (
    ->     SELECT MAX(amt)
    ->     FROM Orders
    -> );
+------+-------------+
| snum | total_sales |
+------+-------------+
| 1001 |    15382.07 |
+------+-------------+
1 row in set (0.012 sec)
```

4. Write a query that selects all customers whose ratings are equal to or greater than ANY of Serres'. (Hint: find all customers of serres salesperson).

```
d2_96712_Devanand> SELECT *
    -> FROM Customers
    -> WHERE rating >= ANY (
    ->     SELECT rating
    ->     FROM Customers
    ->     WHERE snum = (
    ->         SELECT snum
    ->         FROM Salespeople
    ->         WHERE sname = 'Serres'
    ->     )
    -> );
+------+----------+----------+--------+------+
| cnum | cname    | city     | rating | snum |
+------+----------+----------+--------+------+
| 2002 | Giovanni | Rome     |    200 | 1003 |
| 2003 | Liu      | San Jose |    200 | 1002 |
| 2004 | Grass    | Berlin   |    300 | 1002 |
| 2008 | Cisneros | San Jose |    300 | 1007 |
+------+----------+----------+--------+------+
4 rows in set (0.132 sec)
```

5. Write a query that will find all salespeople who have no customers located in their city. (Hint: Choose the correct keyword ANY/ALL).

```
d2_96712_Devanand> SELECT *
    -> FROM Salespeople
    -> WHERE city <> ALL (
    ->     SELECT city
    ->     FROM Customers
    ->     WHERE snum = Salespeople.snum
    -> );
+------+---------+-----------+------+
| snum | sname   | city      | comm |
+------+---------+-----------+------+
| 1004 | Motika  | London    | 0.11 |
| 1007 | Rifkin  | Barcelona | 0.15 |
| 1003 | Axelrod | New York  | 0.10 |
+------+---------+-----------+------+
3 rows in set (0.123 sec)
```

6. Write a query that selects all orders for amounts greater than any for the customers in London.

```
d2_96712_Devanand> SELECT * FROM orders
    -> WHERE amt > ANY (SELECT amt FROM orders WHERE cnum IN
    -> ( SELECT cnum FROM customers WHERE city='London'));
+------+---------+------------+------+------+
| onum | amt     | odate      | cnum | snum |
+------+---------+------------+------+------+
| 3002 | 1900.10 | 1990-10-03 | 2007 | 1004 |
| 3005 | 5160.45 | 1990-10-03 | 2003 | 1002 |
| 3006 | 1098.16 | 1990-10-03 | 2008 | 1007 |
| 3009 | 1713.23 | 1990-10-04 | 2002 | 1003 |
| 3008 | 4723.00 | 1990-10-04 | 2006 | 1001 |
| 3011 | 9891.88 | 1990-10-04 | 2006 | 1001 |
+------+---------+------------+------+------+
6 rows in set (0.453 sec)
```

7. Extract all the orders of Motika.

```
d2_96712_Devanand> SELECT * FROM orders WHERE snum = (SELECT snum
 FROM salespeople WHERE sname='Motika');
+------+---------+------------+------+------+
| onum | amt     | odate      | cnum | snum |
+------+---------+------------+------+------+
| 3002 | 1900.10 | 1990-10-03 | 2007 | 1004 |
+------+---------+------------+------+------+
1 row in set (0.028 sec)
```

8. Find all the orders of salespeople servicing customers in London.

```
d2_96712_Devanand> SELECT *
    -> FROM Orders
    -> WHERE snum IN (
    ->     SELECT DISTINCT snum
    ->     FROM Customers
    ->     WHERE city = 'London'
    -> );
+------+---------+------------+------+------+
| onum | amt     | odate      | cnum | snum |
+------+---------+------------+------+------+
| 3003 |  767.19 | 1990-10-03 | 2001 | 1001 |
| 3008 | 4723.00 | 1990-10-04 | 2006 | 1001 |
| 3011 | 9891.88 | 1990-10-04 | 2006 | 1001 |
+------+---------+------------+------+------+
3 rows in set (0.201 sec)
```

9. Find names and numbers of all salesperson who have more than one customer.

```
d2_96712_Devanand> SELECT snum, sname
    -> FROM Salespeople
    -> WHERE snum IN (
    ->     SELECT snum
    ->     FROM Customers
    ->     GROUP BY snum
    ->     HAVING COUNT(*) > 1
    -> );
+------+--------+
| snum | sname  |
+------+--------+
| 1001 | Peel   |
| 1002 | Serres |
+------+--------+
2 rows in set (0.181 sec)
```

10. Find salespeople number, name and city who have multiple customers. Display the count of customers as well.

```
d2_96712_Devanand> SELECT s.snum, s.sname, s.city, COUNT(c.cnum)
 AS customer_count
    -> FROM Salespeople s
    -> JOIN Customers c ON s.snum = c.snum
    -> GROUP BY s.snum, s.sname, s.city
    -> HAVING COUNT(c.cnum) > 1;
+------+--------+----------+----------------+
| snum | sname  | city     | customer_count |
+------+--------+----------+----------------+
| 1001 | Peel   | London   |              2 |
| 1002 | Serres | San Jose |              2 |
+------+--------+----------+----------------+
2 rows in set (0.143 sec)
```

11. Select customers who have a greater rating than any other customer in Rome.

```
d2_96712_Devanand> SELECT * FROM customers
    -> WHERE rating > ANY(SELECT rating FROM customers
    -> WHERE city NOT IN('Rome'));
+------+----------+----------+--------+------+
| cnum | cname    | city     | rating | snum |
+------+----------+----------+--------+------+
| 2002 | Giovanni | Rome     |    200 | 1003 |
| 2003 | Liu      | San Jose |    200 | 1002 |
| 2004 | Grass    | Berlin   |    300 | 1002 |
| 2008 | Cisneros | San Jose |    300 | 1007 |
+------+----------+----------+--------+------+
4 rows in set (0.137 sec)
```

12. Select all orders that had amounts that were greater that at least one of the orders from ‘1990-10-04’.

```
d2_96712_Devanand> SELECT *
    -> FROM Orders
    -> WHERE amt > ANY (
    ->     SELECT amt
    ->     FROM Orders
    ->     WHERE odate = '1990-10-04'
    -> );
+------+---------+------------+------+------+
| onum | amt     | odate      | cnum | snum |
+------+---------+------------+------+------+
| 3003 |  767.19 | 1990-10-03 | 2001 | 1001 |
| 3002 | 1900.10 | 1990-10-03 | 2007 | 1004 |
| 3005 | 5160.45 | 1990-10-03 | 2003 | 1002 |
| 3006 | 1098.16 | 1990-10-03 | 2008 | 1007 |
| 3009 | 1713.23 | 1990-10-04 | 2002 | 1003 |
| 3008 | 4723.00 | 1990-10-04 | 2006 | 1001 |
| 3010 |  309.95 | 1990-10-04 | 2004 | 1002 |
| 3011 | 9891.88 | 1990-10-04 | 2006 | 1001 |
+------+---------+------------+------+------+
8 rows in set (0.129 sec)
```

13. Find all orders with amounts smaller than any amount for a customer in San Jose.

```
d2_96712_Devanand> SELECT *
    -> FROM Orders
    -> WHERE amt < ANY (
    ->     SELECT amt
    ->     FROM Orders
    ->     WHERE cnum IN (
    ->         SELECT cnum
    ->         FROM Customers
    ->         WHERE city = 'San Jose'
    ->     )
    -> );
+------+---------+------------+------+------+
| onum | amt     | odate      | cnum | snum |
+------+---------+------------+------+------+
| 3001 |   18.69 | 1990-10-03 | 2008 | 1007 |
| 3003 |  767.19 | 1990-10-03 | 2001 | 1001 |
| 3002 | 1900.10 | 1990-10-03 | 2007 | 1004 |
| 3006 | 1098.16 | 1990-10-03 | 2008 | 1007 |
| 3009 | 1713.23 | 1990-10-04 | 2002 | 1003 |
| 3007 |   75.75 | 1990-10-04 | 2004 | 1002 |
| 3008 | 4723.00 | 1990-10-04 | 2006 | 1001 |
| 3010 |  309.95 | 1990-10-04 | 2004 | 1002 |
+------+---------+------------+------+------+
8 rows in set (0.025 sec)
```

14. Select those customers whose rating are higher than every customer in London.

```
d2_96712_Devanand> SELECT * FROM customers
    -> WHERE rating > ALL(SELECT rating FROM customers
    -> WHERE city='London');
+------+----------+----------+--------+------+
| cnum | cname    | city     | rating | snum |
+------+----------+----------+--------+------+
| 2002 | Giovanni | Rome     |    200 | 1003 |
| 2003 | Liu      | San Jose |    200 | 1002 |
| 2004 | Grass    | Berlin   |    300 | 1002 |
| 2008 | Cisneros | San Jose |    300 | 1007 |
+------+----------+----------+--------+------+
4 rows in set (0.014 sec)
```